503 Nuclear Energy

5.3.1 Nuclear fission

Fission splits a heavy nucleus into lighter nuclei, often releasing neutrons and energy. Product kinetic energy can become heat. Nuclear equations conserve charge number and nucleon number, but total rest mass can change; the mass difference corresponds to energy released or absorbed.

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5.3.2 Chain reactions and critical conditions

A chain reaction remains steady when neutrons from each fission cause, on average, one further fission. More than one causes growth and fewer causes decline. Critical mass is the minimum amount needed to sustain the reaction for specified geometry, composition and surroundings, not one universal value for a fuel regardless of conditions.

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5.3.3 Nuclear fusion

Fusion combines light nuclei into a heavier nucleus. Suitable light-nucleus reactions release energy because the products have lower total rest mass. Positive nuclei repel, so very high temperatures and

suitable conditions are needed for close encounters. Fusion and fission change nuclei in opposite ways, but both require consistent mass differences and energy conservation.

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5.3.4 Mass difference and nuclear energy

Total rest-mass difference multiplied by light speed squared gives the corresponding energy change. If reactants have greater total rest mass, the difference can be released as energy. With SI units, use kg to obtain J. Include all products and relevant particles rather than comparing just one nucleus before and after.

$$E_{\text{released}} = \Delta m\, c^2$$

Δm = mass before − mass after (kg), positive for an energy-releasing reaction; E: J; c ≈ 3.00 × 10^8 m s^{-1}.

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5.3.5 The electronvolt

One electronvolt is the energy gained by a particle with one elementary-charge magnitude when accelerated through 1 V. It is an energy unit, not a voltage. Multiply an energy in eV by the elementary charge's numerical SI value to obtain joules. Convert MeV to eV by multiplying by one million before this step.

$$1\ \text{eV} \approx 1.60 \times 10^{-19}\ \text{J}$$

One electronvolt is the kinetic energy gained by an electron accelerated through a potential difference of 1 V. It is an energy unit, not a voltage unit. Rounded textbook conversion.

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5.3.6 · 5.3.7 Atomic mass units and energy conversion

The unified atomic mass unit is one twelfth of the mass of a free, stationary, neutral carbon-12 atom in its ground state. The rest energy associated with 1 u is approximately 931 MeV, so a mass difference in u can be multiplied by 931 to obtain MeV. Mass and energy units are not directly interchangeable without the mass-energy relation.

$$1\ \mathrm{u} = \frac{m(^{12}\mathrm{C})}{12} \approx 1.661 \times 10^{-27}\ \mathrm{kg}$$

$$1\ \mathrm{u} \approx 1.661 \times 10^{-27}\ \mathrm{kg}$$

$$(1\ \mathrm{u})c^2 \approx 931\ \mathrm{MeV}$$

Definition uses a free neutral carbon-12 atom at rest in its ground state; u is the unified atomic mass unit.

Convert nuclear mass from u to kg before using E = mc² in joules. A mass of 1 u corresponds to rest energy of about 931 MeV, or 1 u ≈ 931 MeV/c². Textbook rounded values.

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